4.0 PRE-LAB ASSIGNMENT (3 marks with 1.5 marks for each step):

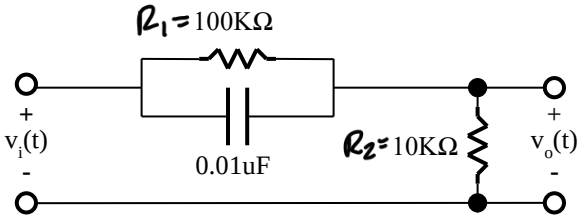


Figure 1.0: An One Pole-One Zero Circuit

(a) Step 1: Consider the network shown in **Figure 1.0**.

i) Derive the transfer function H(s) = V_o/V_i, where V_o is the phasor representation of $v_o(t)$ and V_i is the phasor representation of $v_i(t)$.

Pre-Lab workspace (show your analysis here)

$$Z = \frac{R_1 \cdot \frac{1}{sC}}{R_1 + \frac{1}{sC}} = \frac{R_1}{1+sR_1C}$$

$$H(s) = \frac{V_o}{V_i} = \frac{R_2}{Z+R_2} = \frac{R_2}{\frac{R_1}{1+sR_1C} + R_2} = \frac{R_2(1+sR_1C)}{R_1 + R_2(1+sR_1C)}$$

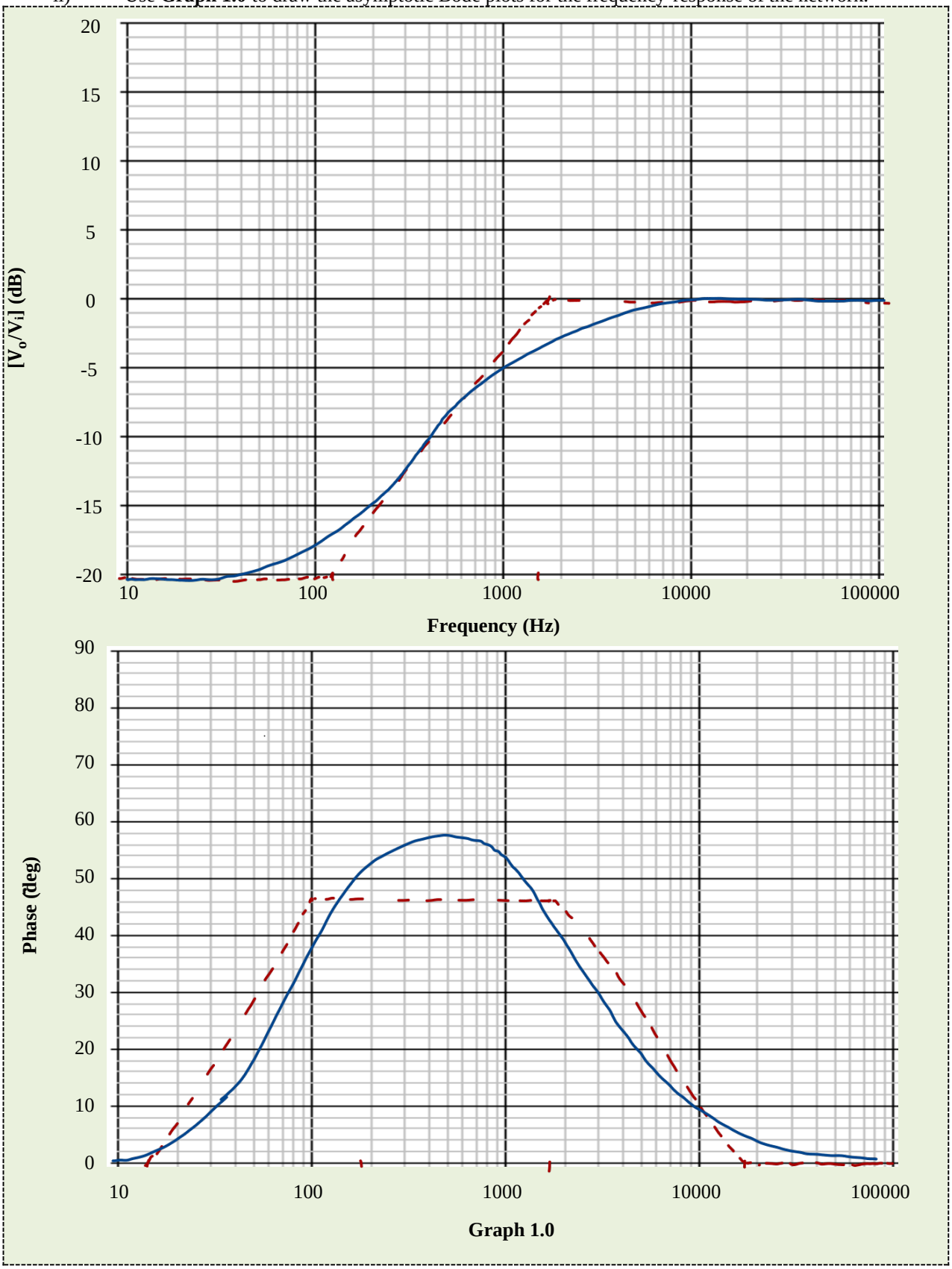
$$= \frac{R_2(1+sR_1C)}{R_1+R_2+sR_1R_2C}$$

$$= \frac{R_2}{R_1+R_2} \cdot \frac{1+sR_1C}{1+s\left(\frac{R_1 \cdot R_2}{R_1+R_2}\right)C}$$

$$= \frac{10K}{100K+10K} \cdot \frac{1+s(100K)C}{1+s\left(\frac{100K \cdot 10K}{100K+10K}\right)C}$$

$$H(s) = 0.91 \cdot \frac{1+s/1K}{1+s/11K}$$

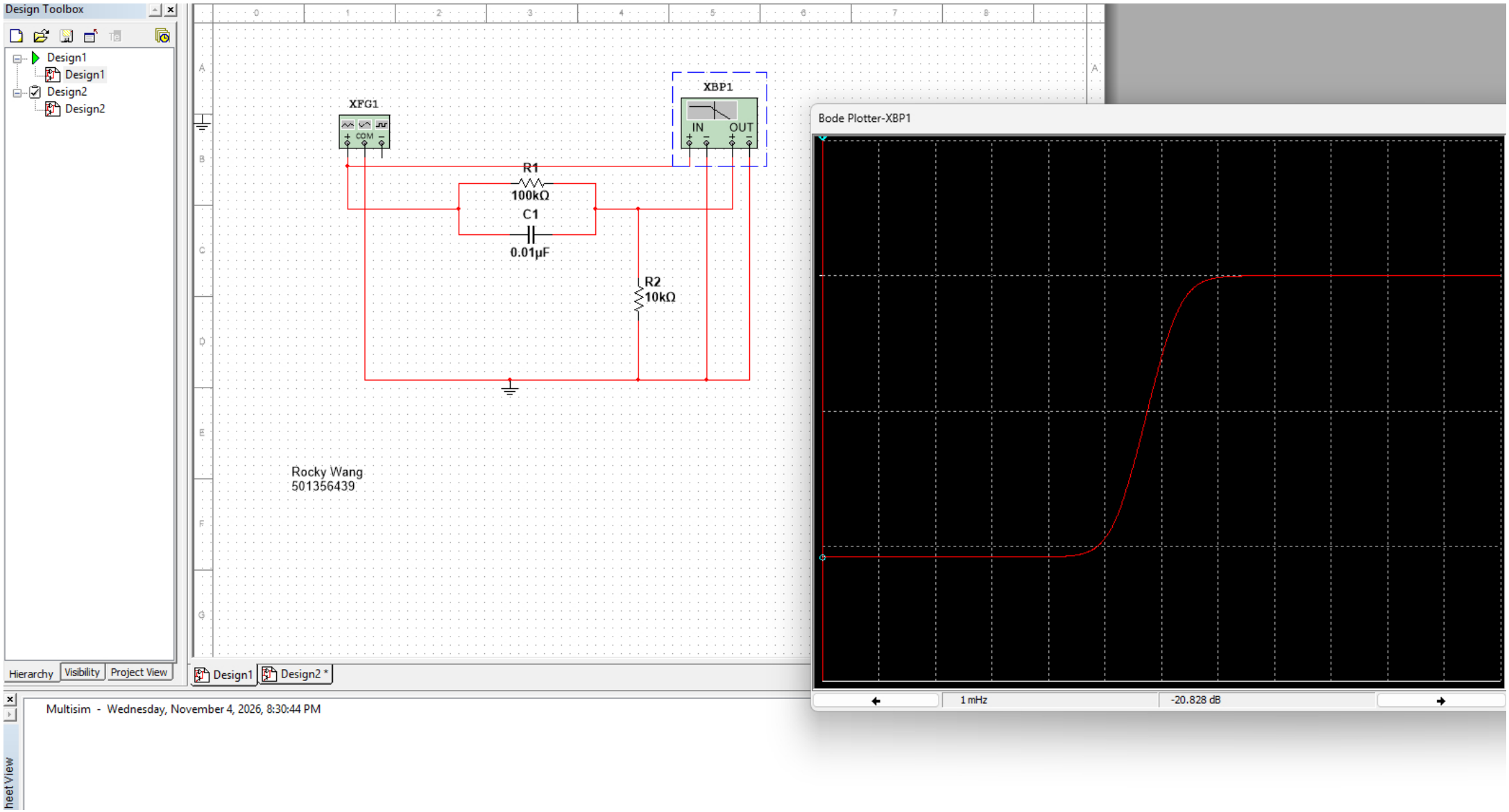
ii) Use **Graph 1.0** to draw the asymptotic Bode plots for the frequency-response of the network.

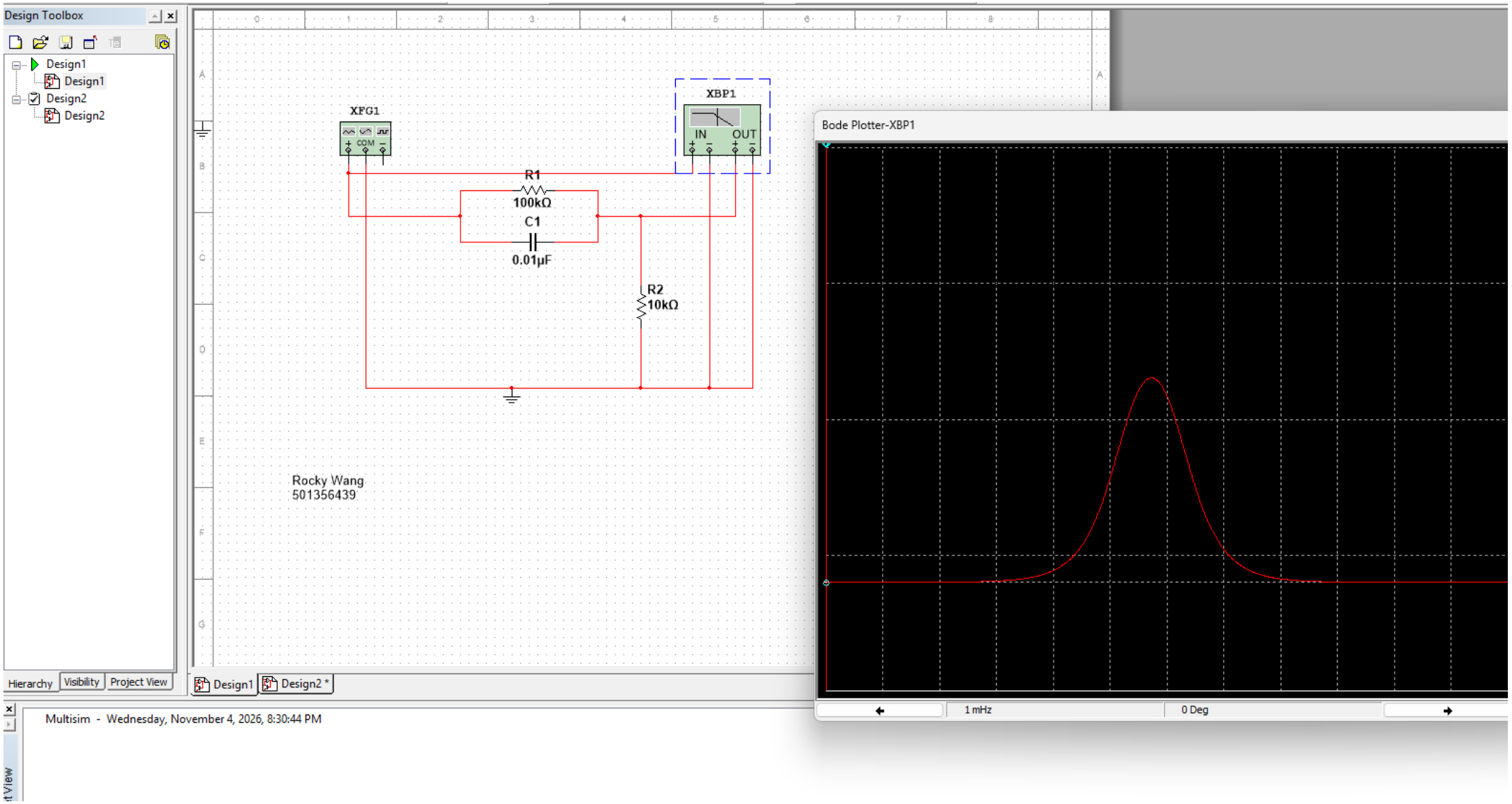


Graph 1.0

iii) Use Multisim to plot the magnitude in dB and phase in degrees of the frequency-responses of the above circuits, for 10Hz ≤ f ≤ 100kHz. (Note: Simulate->Instruments->Bode Plotter can be used in creating the plots).

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
Design2
Design2
XFG1
COM
XBP1
IN
OUT
R1
100kΩ
C1
0.01µF
R2
10kΩ
Rocky Wang
501356439
Bode Plotter-XBP1
1 mHz
-20.828 dB
Hierarchy
Visibility
Project View
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Design Toolbox
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Design2
XFG1
COM
XBP1
IN
OUT
R1
100kΩ
C1
0.01µF
R2
10kΩ
Rocky Wang
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Bode Plotter-XBP1
1 mHz
0 Deg
Hierarchy
Visibility
Project View
Design1
Design2 *
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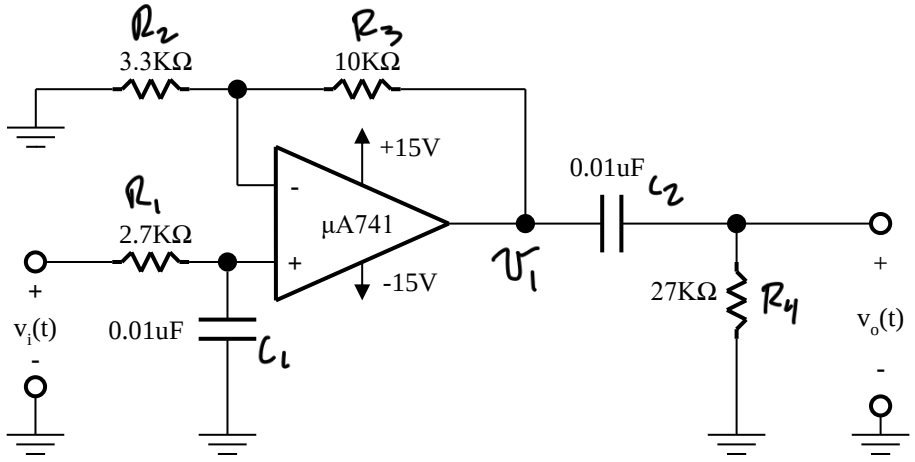


Figure 2.0: A Two Poles-One Zero Circuit

(b) Step 2: Assume that the Op-Amp circuit in **Figure 2.0** is working properly.

i) Derive the transfer function H(s) = V_o/V_i.

Pre-Lab workspace (show your analysis here)

$$V_+ = V_i\left(\frac{Z_{C_1}}{R_1 + Z_{C_1}}\right) = V_i\left(\frac{1/sC}{R_1 + 1/sC}\right) = V_i\left(\frac{1}{1+sR_1C}\right)$$

$$\frac{V_+}{V_i} = \frac{1}{1+sR_1C} = \frac{1}{1+s(27\times10^{-6})}$$

$$V_- = V_1\left(\frac{R_2}{R_2+R_3}\right) = V_+$$

$$\frac{V_+}{V_1} = \frac{R_2+R_3}{R_2} = 4.03\,V$$

$$V_o = V_1\left(\frac{R_4}{R_4+Z_{C_2}}\right) = V_1\left(\frac{R_4}{R_4+1/sC}\right)$$

$$\frac{V_o}{V_1} = \frac{sR_4C}{1+sR_4B} = \frac{s(270\times10^{-6})}{1+s(270\times10^{-6})}$$

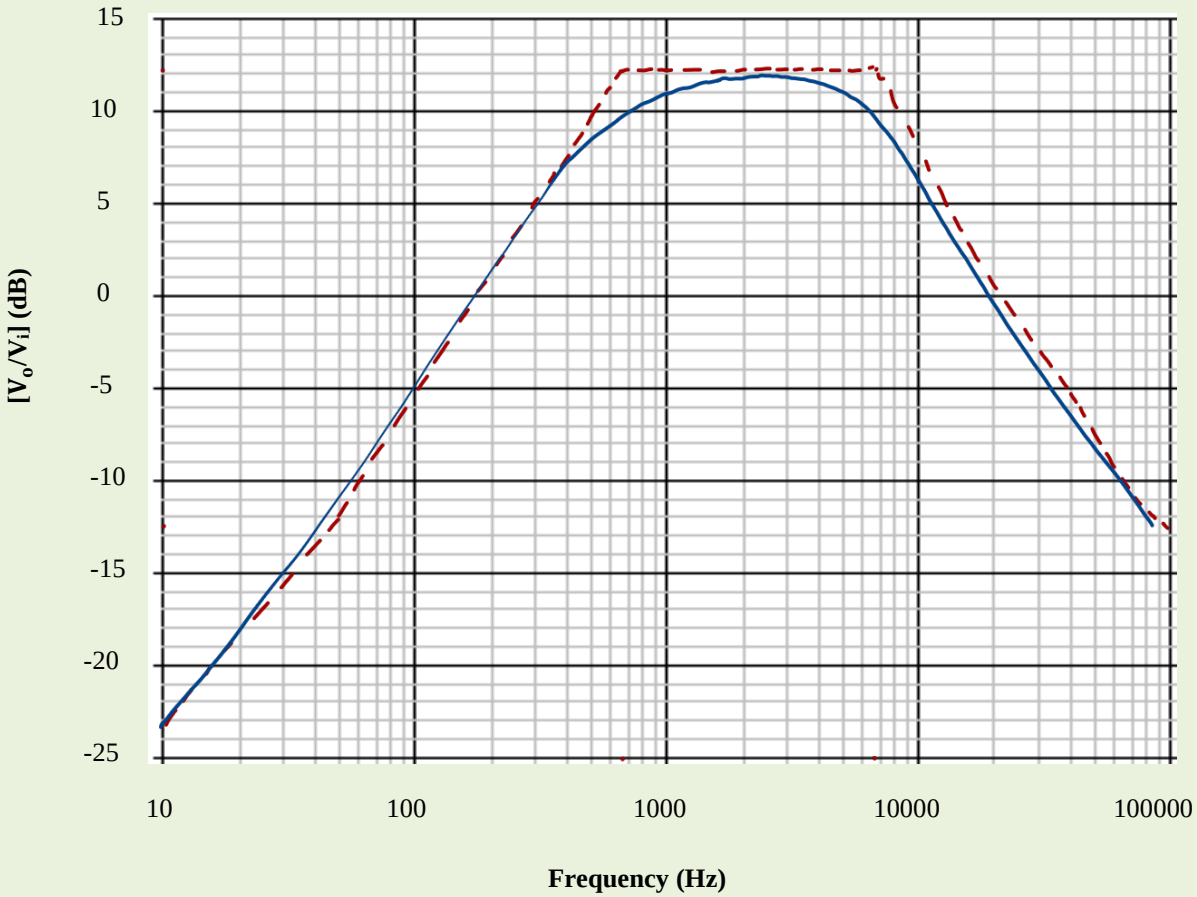
$$H(s) = \frac{V_o}{V_i} = \frac{V_+}{V_i}\cdot\frac{V_1}{V_+}\cdot\frac{V_o}{V_1}$$

$$= \frac{1}{1.s(2.7\times10^{-3})}\cdot 4.03\cdot\frac{s(2.7\times10^{-4})}{1+(2.7\times10^{-4})}$$

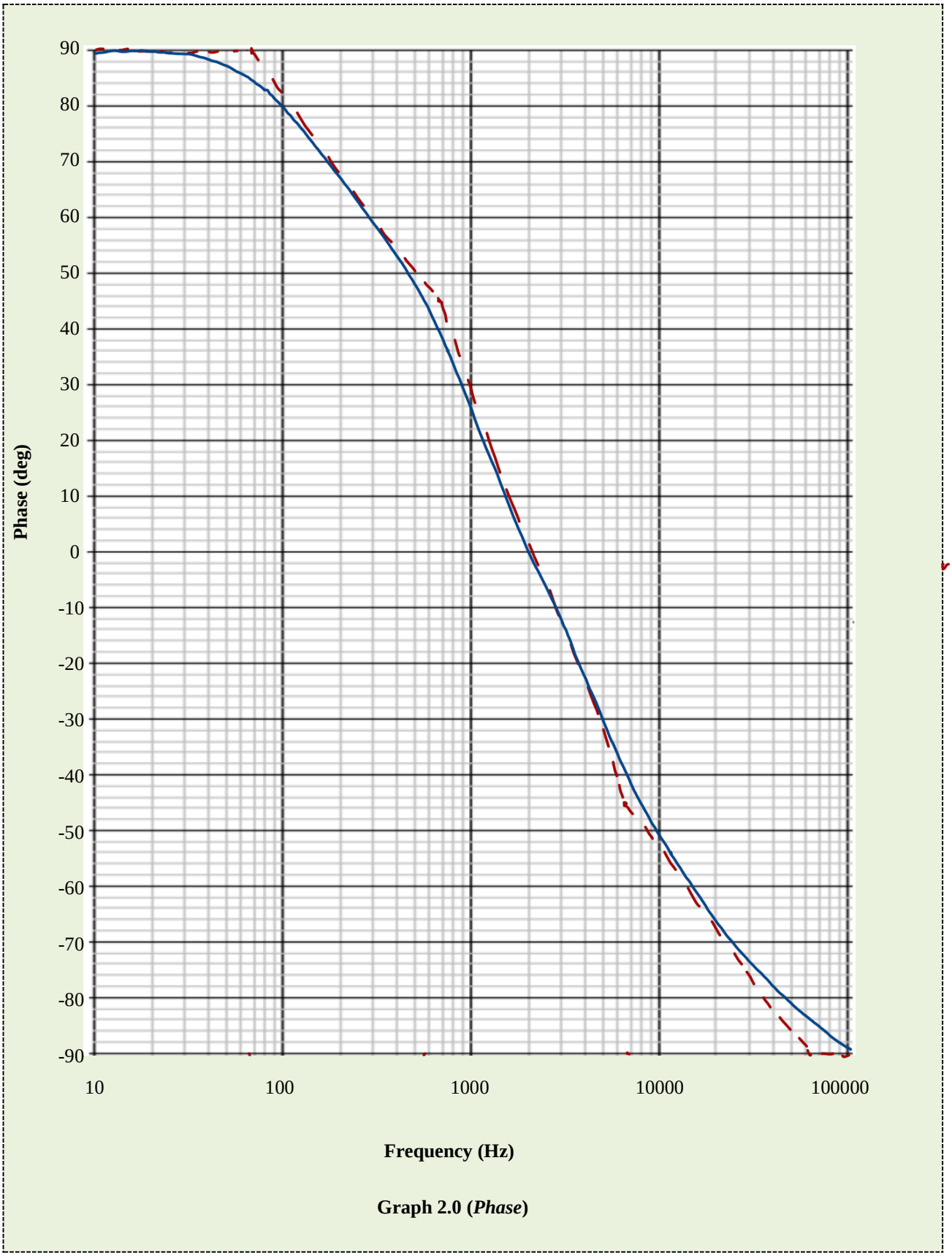
$$= \frac{4.03\,(2.7\times10^{-4})s}{[1+(2.9\times10^{-4})s][1+(2.7\times10^{-5})s]}$$

$$H(s) = 4.03\cdot\frac{s/3704}{(1+s/3704)(1+s/37037)}$$

ii) Use **Graph 2.0** to draw the asymptotic Bode plots for the frequency-response of the circuit:



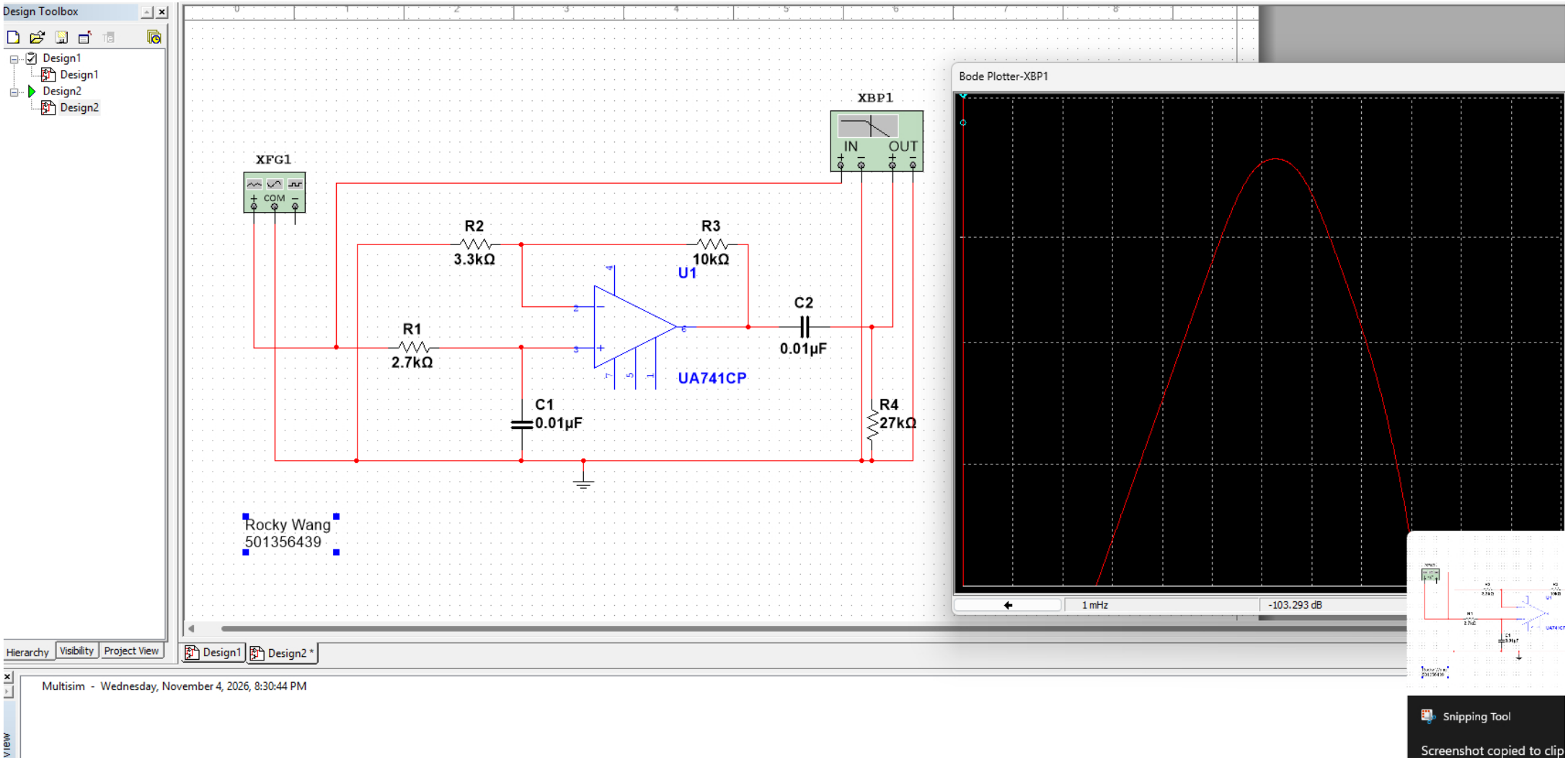
Graph 2.0 (*Gain*)

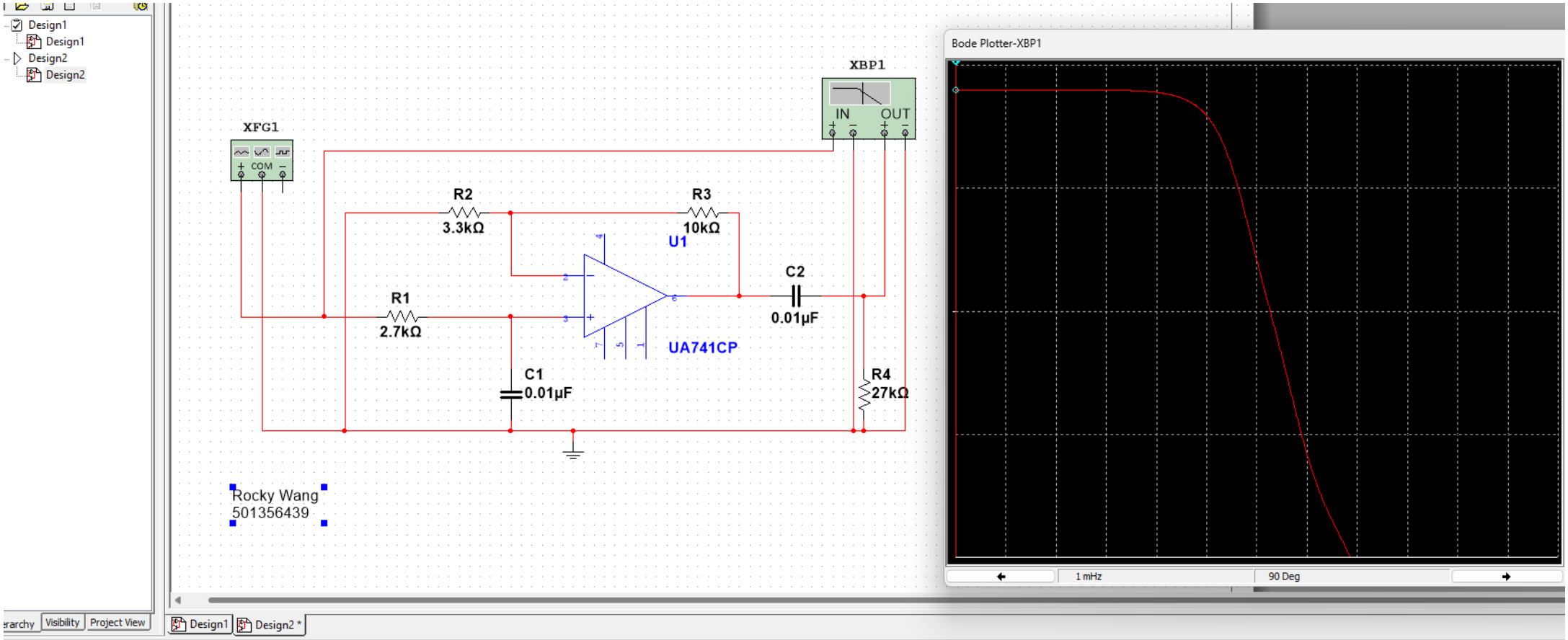


Graph 2.0 (*Phase*)

iii) Use Multisim to plot the magnitude in dB and phase in degrees of the frequency-responses of the above circuits, for $10\text{Hz} \leq f \leq 100\text{kHz}$. (Note: Simulate->Instruments->Bode Plotter can be used in creating the plots).

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
Design2
Design2
XFG1
COM
XBP1
IN
OUT
R2
3.3kΩ
R3
10kΩ
U1
R1
2.7kΩ
C2
0.01µF
UA741CP
C1
0.01µF
R4
27kΩ
Rocky Wang
501356439
Bode Plotter-XBP1
1 mHz
-103.293 dB
Hierarchy
Visibility
Project View
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Design2 *
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Design1
Design1
Design2
Design2
XFG1
COM
XBP1
IN
OUT
R2
3.3kΩ
R3
10kΩ
U1
R1
2.7kΩ
C2
0.01µF
UA741CP
C1
0.01µF
R4
27kΩ
Rocky Wang
501356439
Bode Plotter-XBP1
1 mHz
90 Deg
Visibility
Project View
Design1
Design2 *
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